When Is TSLS Actually LATE? Evidence from Project STAR

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March 2026

GitHub Repository

Abstract. This paper examines whether the conventional LATE interpretation of Two-Stage Least Squares remains valid once covariates are included in the regression specification. Building on Blandhol et al. (2022), it shows that linear IV with covariates is weakly causal only if the rich covariates condition holds, requiring the linear projection of the instrument onto covariates to coincide with its true conditional expectation. The paper reviews the theoretical foundations of this result, presents detailed proofs of the main propositions and theorem, and applies the framework to Project STAR, the Tennessee class-size experiment analyzed by Krueger (1999). The empirical results indicate that standard TSLS estimates fail the Ramsey RESET diagnostic in all grade cohorts, suggesting that the rich covariates condition does not hold in the conventional specification. Flexible alternatives based on Double/Debiased Machine Learning produce lower estimates in several grades, implying nontrivial specification bias in standard TSLS. The findings underscore that in instrumental variables settings, the functional form used to partial out covariates is central to the causal interpretation of the estimator.

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1 Introduction

Two-Stage Least Squares (TSLS) is one of the most widely used tools for estimating causal effects in the presence of endogenous treatments. The interpretation of TSLS is typically grounded in the canonical instrumental variables (IV) framework with a binary instrument and imperfect compliance. In this setting, Imbens and Angrist (1994) show that the TSLS estimand identifies the Local Average Treatment Effect (LATE), defined as the average causal effect for the subpopulation of "compliers".

In empirical practice, however, researchers almost always include covariates in TSLS specifications. As a result, the standard empirical specification differs substantially from the canonical LATE framework. Despite this discrepancy, empirical studies commonly continue to interpret TSLS estimates as LATE without modification.

This raises a fundamental methodological question: under what conditions does the standard TSLS estimator retain a valid LATE interpretation once covariates are included in the regression specification?

An influential methodological contribution by Blandhol et al. (2022) directly addresses this question by examining the mathematical structure of linear IV estimators when covariates are included. Their analysis scrutinizes the implicit mechanics of standard regression implementations of TSLS and clarifies the conditions under which the estimator retains a valid causal interpretation. The paper has had a rapid impact on the empirical methodology literature—reflected in more than 200 Google Scholar citations—highlighting the importance of its challenge to the conventional practice of interpreting TSLS estimates as LATE in specifications that include covariates. Rather than taking the LATE interpretation as given, their framework explicitly characterizes the theoretical and parametric conditions under which this interpretation is justified.

This seminar paper analyzes and extends these insights. Section 2 reviews the canonical LATE framework and the subsequent literature extending these results to settings with covariates. Section 3 analyzes the mathematical decomposition of the TSLS estimand when covariates are introduced, formalizing the concepts of weakly causal estimands and rich covariates. Section 4 provides a critical discussion of the theoretical requirements underlying these conditions and their practical implications for applied research. Finally, Section 5 applies these theoretical insights to the Tennessee STAR experiment originally analyzed by Krueger (1999), employing the Ramsey RESET test to evaluate whether the empirical specification satisfies the conditions required for a valid causal interpretation.

2 Literature Review

2.1 The LATE Framework Without Covariates

To place the methodological issues identified by Blandhol et al. (2022) in context, it is helpful to first review the canonical identification results for IV estimators without covariates. The potential outcomes framework developed by Imbens and Angrist (1994) provides the foundation for the standard interpretation of these estimators.

Following Imbens and Angrist (1994), let $Y_i(0)$ be the potential outcome for individual i without the treatment, and $Y_i(1)$ be the potential outcome with the treatment. The observed treatment indicator is $D_i \in \{0, 1\}$, making the observed outcome $Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$. The individual causal effect is $Y_i(1) - Y_i(0)$, which is fundamentally unobservable for any single individual.

To identify the average treatment effect in observational studies, researchers rely on an instrumental variable, Z_i. Let $\mathfrak{Z}$ be the support of Z_i. For each $z \in \mathfrak{Z}$, define a potential treatment indicator $D_i(z)$, representing the treatment status individual i would select if assigned the instrument value $Z_i = z$. The identification strategy relies on two core conditions provided by Imbens and Angrist (1994). The first ensures the instrument is exogenous and relevant:

> **Condition 1 (Existence of Instruments):** Let Z_i be a random variable such that (i) for all $w \in \mathfrak{Z}$, the potential outcomes and potential treatment assignments $(Y_i(0), Y_i(1), D_i(w))$ are jointly independent of Z_i, and (ii) the probability of treatment, $P(w) = E[D_i | Z_i = w]$, is a nontrivial function of w.

Condition 1 alone is insufficient to identify an average treatment effect. To understand why, we follow the derivation of Imbens and Angrist (1994) and examine the expected difference in outcomes for two points of support of the instrument, z and w, assuming $P(z) > P(w)$. Using the independence assumption, this difference expands to:

$$\begin{aligned} E[Y_i|Z_i = z] - E[Y_i|Z_i = w] &= E[D_i(z) \cdot Y_i(1) + (1 - D_i(z)) \cdot Y_i(0)|Z_i = z] \\ &\quad - E[D_i(w) \cdot Y_i(1) + (1 - D_i(w)) \cdot Y_i(0)|Z_i = w] \\ &= E[(D_i(z) - D_i(w)) \cdot (Y_i(1) - Y_i(0))] \end{aligned}$$

By decomposing the expectation across the subgroups whose treatment status changes, we get:

$$\begin{aligned} &E[Y_i \mid Z_i = z] - E[Y_i \mid Z_i = w] \\ &= Pr[D_i(z) - D_i(w) = 1] \cdot E[Y_i(1) - Y_i(0) \mid D_i(z) - D_i(w) = 1] \\ &\quad - Pr[D_i(z) - D_i(w) = -1] \cdot E[Y_i(1) - Y_i(0) \mid D_i(z) - D_i(w) = -1] \end{aligned} \tag{1}$$

Equation 1 reveals the fundamental identification problem: the difference in expected outcomes is a mixture of treatment effects for "compliers" ($D_i(z) - D_i(w) = 1$) and "defiers" ($D_i(z) - D_i(w) = -1$). Because the defiers shift out of treatment when the instrument induces participation, their treatment effect receives a negative weight. Consequently, the overall estimand could be zero or negative even if the true causal effect is strictly positive for all individuals.

To prevent this negative weighting, Imbens and Angrist (1994) introduced their second core assumption:

> **Condition 2 (Monotonicity):** For all $z, w \in \mathfrak{Z}$, either $D_i(z) \geq D_i(w)$ for all i, or $D_i(z) \leq D_i(w)$ for all i.

Monotonicity ensures that the instrument affects the selection decision in only one direction, completely ruling out the existence of defiers. By assuming $D_i(z) \geq D_i(w)$, the probability $Pr[D_i(z) - D_i(w) = -1] = 0$, which zeroes out the second term in Equation 1. This simplifies the relationship to:

$$E[Y_i|Z_i = z] - E[Y_i|Z_i = w] = (P(z) - P(w)) \cdot E[Y_i(1) - Y_i(0)|D_i(z) - D_i(w) = 1]$$

Dividing both sides by $P(z) - P(w)$ yields the classic Wald estimator:

$$\alpha_{z,w} = \frac{E[Y_i \mid Z_i = z] - E[Y_i \mid Z_i = w]}{P(z) - P(w)} = E[Y_i(1) - Y_i(0) \mid D_i(z) \neq D_i(w)]\,.$$

This result establishes the main identification theorem of Imbens and Angrist (1994).

Theorem 1 (Imbens and Angrist, 1994). If Conditions 1 and 2 hold, then the following average treatment effect can be identified:

$$\alpha_{z,w} = \mathbb{E}\left[Y_i(1) - Y_i(0) \mid D_i(z) \neq D_i(w)\right]. \tag{2}$$

This parameter is identified from the joint distribution of Y, D, and Z for all z and w in the support of Z such that $\mathbb{E}[Y_i \mid Z_i = z]$ and $\mathbb{E}[Y_i \mid Z_i = w]$ are finite and $P(z) \neq P(w)$.

Ultimately, this derivation proves that in the absence of covariates, the IV estimator achieves a clean causal interpretation. It isolates the average treatment effect for the "compliers", and because monotonicity rules out defiers, it guarantees strictly non-negative weights.

2.2 Extensions to Covariates and Modern Critiques

While the derivation by Imbens and Angrist (1994) provided a causal interpretation for the simple IV estimator, empirical applications almost universally require the inclusion

of covariates.

As the use of covariates became standard practice, a conventional wisdom emerged that the clean LATE interpretation naturally extended to these more complex models. This view was prominently championed by Angrist and Pischke (2009), who asserted that TSLS with covariates produces an average of covariate-specific LATEs, thereby providing a "simple causal interpretation for 2SLS in most empirically relevant settings". However, this theoretical guarantee was based on a "saturate and weight" approach, which requires the model to control for covariates nonparametrically (i.e., using a fully saturated model with indicators for all possible covariate combinations). In practice, empirical researchers rarely saturate their models due to the "data-hungry" nature of the approach and the risk of many-instruments bias (Blandhol et al., 2022).

In recent years, a growing body of econometric literature has begun to scrutinize the implicit weighting mechanisms of standard regression estimators. A prominent example is the recent wave of difference-in-differences literature, such as Goodman-Bacon (2021), which demonstrated that two-way fixed effects (TWFE) models can yield severely biased estimates when treatment effects are heterogeneous, due to the presence of negative weights. While the TWFE critiques focus on treatment effect heterogeneity across time or cohorts, the problems identified in the IV context are arguably more fundamental. As Blandhol et al. (2022) note, the issues with TSLS weights persist even if one assumes that treatment effects are perfectly constant.

Within the specific domain of instrumental variables, recent methodological papers have investigated the conditions under which TSLS retains a valid causal interpretation. Kolesár (2013) provided general sufficient conditions for the TSLS estimand to equal a non-negatively weighted average of LATEs. However, his mathematical proofs explicitly maintained a "rich covariates" assumption—requiring that the linear projection of the instrument onto the covariates perfectly matches its true conditional mean. Similarly, Słoczyński (2024) offered a severe critique of the LATE interpretation of TSLS, focusing on cases where the direction of monotonicity varies across covariate groups. He demonstrated that unless the first-stage equation includes instrument-covariate interactions, the TSLS estimand can fail to be weakly causal. Yet, like Kolesár, Słoczyński (2024) still maintained the assumption of rich covariates in his baseline analysis.

Blandhol et al. (2022) bridge this gap and push the literature forward by challenging the rich covariates assumption itself. They demonstrate that for the vast majority of empirical papers that use parametric (non-saturated) controls, the rich covariates assumption is routinely violated. By focusing on strict, unconditional monotonicity, Blandhol et al. (2022) prove that the rich covariates condition is not merely a sufficient assumption, but a strictly *necessary* one for TSLS to have any causal interpretation. Without it, the TSLS estimand suffers from "level dependence," meaning the estimate is contaminated by the baseline outcome levels of always-takers and never-takers, inevitably applying

negative weights to some subgroups.

Together, this modern literature reveals that the conventional extrapolation of the Imbens and Angrist (1994) LATE framework to parametric TSLS models is mathematically fragile.

3 Core Methodology: When is TSLS Actually LATE?

This section details the central methodological contribution of Blandhol et al. (2022). We analyze the standard instrumental variables framework under the inclusion of covariates in the special case of a binary treatment and a binary instrument.

3.1 Binary IV with Covariates Setup

Following Blandhol et al. (2022), let $T \in \{0, 1\}$ be a binary treatment, $Z \in \{0, 1\}$ a binary instrument, Y the observed outcome, and X a vector of covariates that includes a constant. The potential outcomes are $Y(0)$ and $Y(1)$, and the potential treatment choices are $T(0)$ and $T(1)$. We maintain the standard IV assumptions conditional on covariates:

1. **Conditional Exogeneity:** $(Y(0), Y(1), T(0), T(1)) \perp Z \mid X$.
2. **Monotonicity:** $Pr[T(1) \geq T(0)] = 1$.

Monotonicity restricts the population to three compliance groups $G \in \{NT, CP, AT\}$: Never-Takers ($T(0) = 0, T(1) = 0$), Compliers ($T(0) = 0, T(1) = 1$), and Always-Takers ($T(0) = 1, T(1) = 1$).

By the Frisch-Waugh-Lovell theorem, the linear IV (or TSLS) estimand for the coefficient on T with covariates X is given by:

$$\beta_{IV} = \frac{\mathbb{E}[Y\tilde{Z}]}{\mathbb{E}[T\tilde{Z}]}$$

where $\tilde{Z} \equiv Z - \mathbb{L}[Z|X]$ are the residuals from the linear projection of Z onto X, and $\mathbb{L}[Z|X] \equiv X'\mathbb{E}[XX']^{-1}\mathbb{E}[XZ]$ are the population fitted values.

3.2 Propositions 1 and 2

Conventional wisdom suggests that β_{IV} represents a non-negatively weighted average of treatment effects strictly for the compliers. Proposition 1 demonstrates that this is generally false when covariates are present.

Proposition 1. Suppose $\mathbb{E}[Y(t)|X] = \eta_t' X$ for $t \in \{0, 1\}$. Let $\Delta(G, x) \equiv \mathbb{E}[Y(1) - Y(0) \mid G, X = x]$ denote the conditional average treatment effect for group G. Then

(Blandhol et al., 2022):

$$\beta_{IV} = \mathbb{E}[\omega(CP, X)\Delta(CP, X)] + \mathbb{E}[\omega(AT, X)\Delta(AT, X)]$$

$$\text{where} \quad \omega(\text{CP}, X) \equiv \mathbb{E}[Z \mid X]\big(1 - \mathbb{L}[Z \mid X]\big)\Pr(G = \text{CP} \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1},$$
$$\omega(\text{AT}, X) \equiv \mathbb{E}[\tilde{Z} \mid X]\Pr(G = \text{AT} \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1}.$$

Sign properties: If $\mathbb{E}[\tilde{Z}T] > 0$, then the complier weights $\omega(CP, X)$ are negative if and only if $\mathbb{L}[Z|X] > 1$. The always-taker weights $\omega(AT, X)$ are strictly negative with positive probability unless $\mathbb{E}[\tilde{Z}|X] = 0$ deterministically.

Proposition 1 demonstrates that, in general, β_{IV} reflects the treatment effects of not only the compliers, but also the always-takers. Furthermore, the sign properties of Proposition 1 show that the TSLS estimand is not generally a non-negatively weighted average of causal effects.

Always-Taker Weights. Because $Pr(G = AT|X)$ is a probability, it is non-negative. Under the monotonicity assumption, the first-stage coefficient is positive ($\mathbb{E}[\tilde{Z}T] > 0$). Thus, the sign of the always-taker weight, $\omega(AT, X)$, depends on the projection error, $\mathbb{E}[\tilde{Z}|X] = \mathbb{E}[Z|X] - \mathbb{L}[Z|X]$. Because the covariate vector X includes a constant, the unconditional expectation of this error is zero ($\mathbb{E}[\mathbb{E}[\tilde{Z}|X]] = 0$). Unless the linear projection equals the conditional mean for all observations ($\mathbb{E}[\tilde{Z}|X] = 0$ deterministically), the projection error takes negative values with positive probability. Consequently, non-saturated TSLS specifications assign negative weights to the treatment effects of always-takers, violating even a minimal condition for being causal.

Complier Weights. Proposition 1 also establishes that complier weights, $\omega(CP, X)$, can be negative. Similar to the always-takers, the probability $Pr(G = CP|X)$ and the first-stage denominator $\mathbb{E}[\tilde{Z}T]$ are strictly non-negative. Furthermore, because the instrument Z is binary, its true conditional expectation $\mathbb{E}[Z|X]$ is bounded between 0 and 1, meaning it is also non-negative. Therefore, the weight is proportional to $(1 - \mathbb{L}[Z|X])$, meaning compliers receive negative weights when the linear projection yields fitted values exceeding one ($\mathbb{L}[Z|X] > 1$). This occurs when a binary instrument is modeled using a Linear Probability Model with continuous or non-saturated covariates, which does not constrain predictions to the $[0, 1]$ interval.

To avoid negative weights for compliers and always-takers, it is necessary that $\mathbb{L}[Z|X] = \mathbb{E}[Z|X]$, a requirement defined as "rich covariates".

This decomposition is not unique. As Proposition 2 shows, β_{IV} admits a continuum of valid decompositions indexed by a parameter ϵ, all of which introduce negative weights on either always-takers, never-takers, or both.

Proposition 2. Under the same assumptions as Proposition 1, for any real number ϵ (Blandhol et al., 2022):

$$\beta_{IV} = \mathbb{E}[\omega_\epsilon(CP, X)\Delta(CP, X)] + \mathbb{E}[\omega_\epsilon(AT, X)\Delta(AT, X)] + \mathbb{E}[\omega_\epsilon(NT, X)\Delta(NT, X)]$$

$$\begin{aligned}
\text{where} \quad & \omega_\epsilon(\mathrm{CP}, X) \equiv \Big(\epsilon\,\mathbb{E}[\tilde{Z} \mid X] + \mathbb{L}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big)\Big)\,\mathbb{P}[G = \mathrm{CP} \mid X]\,\mathbb{E}[\tilde{Z}T]^{-1}, \\
& \omega_\epsilon(\mathrm{AT}, X) \equiv \epsilon\,\mathbb{E}[\tilde{Z} \mid X]\,\mathbb{P}[G = \mathrm{AT} \mid X]\,\mathbb{E}[\tilde{Z}T]^{-1}, \\
\text{and} \quad & \omega_\epsilon(\mathrm{NT}, X) \equiv (\epsilon - 1)\,\mathbb{E}[\tilde{Z} \mid X]\,\mathbb{P}[G = \mathrm{NT} \mid X]\,\mathbb{E}[\tilde{Z}T]^{-1}.
\end{aligned}$$

The scalar ϵ dictates how the bias induced by the projection error, $\mathbb{E}[\tilde{Z}|X]$, is mathematically allocated between always-takers and never-takers. Setting $\epsilon = 1$ recovers Proposition 1 by assigning all projection bias to the always-takers, whereas setting $\epsilon = 0$ shifts this bias entirely to the never-takers. Because the unconditional expectation of the projection error is zero ($\mathbb{E}[\mathbb{E}[\tilde{Z}|X]] = 0$), the error must take both positive and negative values unless $\mathbb{E}[\tilde{Z}|X] = 0$ deterministically. Consequently, for any real value of ϵ, either the always-taker weights, the never-taker weights, or both must take strictly negative values with positive probability. This proves that no decomposition exists that bounds all weights at or above zero, reinforcing the conclusion that the TSLS estimand cannot be interpreted as a non-negatively weighted average of causal effects without the rich covariates assumption.

Definition RC (Rich Covariates). Let $\mathbb{L}[Z|X = x] \equiv x'\mathbb{E}[XX']^{-1}\mathbb{E}[XZ]$ denote the population fitted value at $X = x$ from the linear regression of Z onto X. An IV specification is said to have rich covariates if (Blandhol et al., 2022):

$$\mathbb{E}[Z|X = x] = \mathbb{L}[Z|X = x]$$

for every $x \in \mathcal{X}$.

The rich covariates condition requires that the linear projection of the instrument onto the covariates exactly reproduces the true conditional expectation of the instrument. In other words, the first-stage regression of Z on X must be correctly specified so that the projection error $\mathbb{E}[\tilde{Z}|X]$ is identically zero. When this condition holds, the residualized instrument $\tilde{Z}$ is orthogonal to all functions of the covariates, eliminating the projection bias that generates negative weights in Propositions 1 and 2.

However, in most empirical applications the conditional expectation $\mathbb{E}[Z|X]$ is unlikely to be exactly linear in X, especially when covariates are continuous or when the specification omits interactions or nonlinear terms. Consequently, the rich covariates condition is a strong restriction and is rarely satisfied in practice.

3.3 Proof of Proposition 2

To understand exactly how these negative weights emerge and why the standard TSLS estimand suffers from level dependence, we must walk through the mathematical proof of Proposition 2. The following proofs closely follow the logic of Blandhol et al. (2022), but expand the intermediate steps for clarity.

Step 1: Decomposing the Numerator. We begin by decomposing the numerator of the IV estimand, $\mathbb{E}[Y\tilde{Z}]$, using the Law of Iterated Expectations. Conditioning on the covariates X yields

$$\mathbb{E}[Y\tilde{Z}] = \mathbb{E}\big[\mathbb{E}[Y\tilde{Z} \mid X]\big].$$

Recall that $\tilde{Z} \equiv Z - \mathbb{L}[Z \mid X]$, where $\mathbb{L}[Z \mid X]$ is the linear projection of Z onto X and is therefore a deterministic function of X. Using the covariance identity $\mathbb{E}[AB] = Cov(A, B) + \mathbb{E}[A]\mathbb{E}[B]$, the inner expectation can be written as

$$\mathbb{E}[Y\tilde{Z} \mid X] = Cov(Y, \tilde{Z} \mid X) + \mathbb{E}[Y \mid X]\mathbb{E}[\tilde{Z} \mid X].$$

Because $\mathbb{L}[Z \mid X]$ is deterministic conditional on X, we have $Cov(Y, \tilde{Z} \mid X) = Cov(Y, Z \mid X)$. Substituting this expression back into the outer expectation yields the following decomposition of the numerator:

$$\mathbb{E}[Y\tilde{Z}] = \mathbb{E}\big[Cov(Y, Z \mid X)\big] + \mathbb{E}\big[\mathbb{E}[Y \mid X]\mathbb{E}[\tilde{Z} \mid X]\big]. \tag{3}$$

Step 2: The Conditional Covariance Term. The first term in Equation 3, $\mathbb{E}[Cov(Y, Z|X)]$, represents the numerator of a fully nonparametric IV estimator that conditions flexibly on X. Applying the conditional LATE identification argument of Imbens and Angrist (1994)—that is, assuming conditional independence and monotonicity given X—variation in the instrument affects treatment decisions only for the complier group. Consequently, the covariance between the outcome and the instrument conditional on X is driven solely by the treatment effect for compliers:

$$Cov(Y, Z|X) = \Delta(CP, X)Cov(T, Z|X).$$

Since T and Z are binary, the conditional covariance of treatment and the instrument equals the probability of compliance multiplied by the variance of the instrument:

$$Cov(T, Z|X) = Pr(G = CP|X)\mathbb{E}[Z|X]\big(1 - \mathbb{E}[Z|X]\big).$$

Substituting this expression into the previous equation yields

$$\mathrm{Cov}(Y, Z \mid X) = \Delta(\mathrm{CP}, X)\,\Pr(G = \mathrm{CP} \mid X)\,\underbrace{\mathbb{E}[Z \mid X]\,(1 - \mathbb{E}[Z \mid X])}_{\mathrm{Var}(Z|X)}.$$

Thus, the conditional covariance term depends only on the treatment effect for compliers and the variance of the instrument. Consequently, this component of the numerator captures purely causal variation and does not introduce bias. Any deviation from the LATE interpretation must therefore arise from the second term in Equation 3.

Step 3: Expanding the Conditional Mean of the Outcome. The second term, $\mathbb{E}[\mathbb{E}[Y|X]\mathbb{E}[\tilde{Z}|X]]$, captures the discrepancy between nonparametric conditioning and linear projection. To evaluate it, we expand $\mathbb{E}[Y|X]$ using the law of total expectation over the unobserved compliance types:

$$\begin{aligned}\mathbb{E}[Y|X] = \mathbb{E}[Y|G = AT, X]Pr(G = AT|X) \\ + \mathbb{E}[Y|G = NT, X]Pr(G = NT|X) \\ + \mathbb{E}[Y|G = CP, X]Pr(G = CP|X).\end{aligned}$$

We now substitute the potential outcomes for each compliance group. Under the conditional independence assumption, the instrument does not affect potential outcomes directly. Always-takers therefore always receive the treated outcome $Y(1)$, never-takers always receive the untreated outcome $Y(0)$, and compliers receive $(1 - Z)Y(0) + ZY(1)$. This yields

$$\begin{aligned}\mathbb{E}[Y|X] = \mathbb{E}[Y(1)|G = AT, X]Pr(G = AT|X) \\ + \mathbb{E}[Y(0)|G = NT, X]Pr(G = NT|X) \\ + \mathbb{E}[(1 - Z)Y(0) + ZY(1)|G = CP, X]Pr(G = CP|X). \qquad (4)\end{aligned}$$

Step 4: The Centering Argument. To isolate treatment effects, Equation 4 must be centered by subtracting covariate-specific baseline outcomes. Because always-takers and never-takers deterministically reveal $Y(1)$ and $Y(0)$ respectively, we can center the conditional mean around either the untreated baseline, $\mathbb{E}[Y(0)|X]$, or the treated baseline, $\mathbb{E}[Y(1)|X]$.

First, we can add and subtract the untreated baseline for the always-takers: $\mathbb{E}[Y(0)|G = AT, X]Pr(G = AT|X)$. Using the assumption of linearity in potential outcomes, $\mathbb{E}[Y(0)|X] = \eta_0' X$, this centering removes the average untreated outcome for the never-takers, leaving only the treatment effects for compliers and always-takers plus the linear untreated base-

line.

$$
\begin{aligned}
\mathbb{E}[Y|X] &= \mathbb{E}[Y(1)|G=AT,X]Pr(G=AT|X) + \mathbb{E}[Y(0)|G=NT,X]Pr(G=NT|X) \\
&\quad + \mathbb{E}\big[Y(0) + Z\big(Y(1)-Y(0)\big) \mid G=CP,X\big]Pr(G=CP|X) \\
&= \mathbb{E}[Y(1)|G=AT,X]Pr(G=AT|X) + \mathbb{E}[Y(0)|G=NT,X]Pr(G=NT|X) \\
&\quad + \mathbb{E}[Y(0)|G=CP,X]Pr(G=CP|X) + \Delta(CP,X)Pr(G=CP|X)\mathbb{E}[Z|X] \\
&= \Big(\mathbb{E}[Y(1)|G=AT,X] - \mathbb{E}[Y(0)|G=AT,X]\Big)Pr(G=AT|X) \\
&\quad + \Big(\mathbb{E}[Y(0)|G=AT,X]Pr(G=AT|X) + \mathbb{E}[Y(0)|G=NT,X]Pr(G=NT|X) \\
&\qquad + \mathbb{E}[Y(0)|G=CP,X]Pr(G=CP|X)\Big) + \Delta(CP,X)Pr(G=CP|X)\mathbb{E}[Z|X] \\
&= \Delta(AT,X)Pr(G=AT|X) + \mathbb{E}[Y(0)|X] + \Delta(CP,X)Pr(G=CP|X)\mathbb{E}[Z|X] \\
&= \Delta(CP,X)Pr(G=CP|X)\mathbb{E}[Z|X] + \Delta(AT,X)Pr(G=AT|X) + \eta_0'X. \qquad (5)
\end{aligned}
$$

where by the Law of Total Expectation:

$$
\begin{aligned}
\mathbb{E}[Y(0) \mid X] &= \mathbb{E}[Y(0) \mid G=AT,X]Pr(G=AT \mid X) + \mathbb{E}[Y(0) \mid G=NT,X]Pr(G=NT \mid X) \\
&\quad + \mathbb{E}[Y(0) \mid G=CP,X]Pr(G=CP \mid X)
\end{aligned}
$$

Alternatively, we could center around the treated baseline $\mathbb{E}[Y(1)|X] = \eta_1'X$ by adding and subtracting $\mathbb{E}[Y(1)|G=NT,X]Pr(G=NT|X)$. This isolates never-takers instead of always-takers:

$$
\mathbb{E}[Y|X] = \Delta(CP,X)Pr(G=CP|X)(\mathbb{E}[Z|X]-1) - \Delta(NT,X)Pr(G=NT|X) + \eta_1'X \quad (6)
$$

To capture the full continuum of valid decompositions, we multiply Equation 5 by an arbitrary real number ϵ and Equation 6 by $(1-\epsilon)$, and sum them together. This produces a generalized expression for $\mathbb{E}[Y|X]$:

$$
\begin{aligned}
\mathbb{E}[Y|X] &= \Delta(CP,X)Pr(G=CP|X)(\mathbb{E}[Z|X] + \epsilon - 1) \\
&\quad + \Delta(AT,X)\epsilon Pr(G=AT|X) + \Delta(NT,X)(\epsilon-1)Pr(G=NT|X) \\
&\quad + \epsilon\eta_0'X + (1-\epsilon)\eta_1'X \qquad (7)
\end{aligned}
$$

Step 5: Orthogonality and the Final Weights. We now substitute Equation 7 back into the second term of Equation 3, $\mathbb{E}[\mathbb{E}[Y|X]\mathbb{E}[\tilde{Z}|X]]$. Recall that $\tilde{Z}$ is the residual from the linear projection of Z onto X. By construction, this residual is orthogonal to all linear combinations of X, implying $\mathbb{E}[X\tilde{Z}] = 0$.

Consequently, when Equation 7 is multiplied by $\mathbb{E}[\tilde{Z}|X]$ and the outer expectation is taken, the baseline terms involving $\eta_0'X$ and $\eta_1'X$ vanish due to this orthogonality

condition:

$$
\begin{aligned}
\mathbb{E}\big[\mathbb{E}[Y|X]\mathbb{E}[\tilde{Z}|X]\big] &= \mathbb{E}\Big[\Delta(CP,X)Pr(G=CP|X)(\mathbb{E}[Z|X]+\epsilon-1)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(AT,X)\epsilon Pr(G=AT|X)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(NT,X)(\epsilon-1)Pr(G=NT|X)\mathbb{E}[\tilde{Z}|X]\Big] \\
&\quad + \mathbb{E}\Big[\big(\epsilon\eta_0'X+(1-\epsilon)\eta_1'X\big)\mathbb{E}[\tilde{Z}|X]\Big] \\
&= \mathbb{E}\Big[\Delta(CP,X)Pr(G=CP|X)(\mathbb{E}[Z|X]+\epsilon-1)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(AT,X)\epsilon Pr(G=AT|X)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(NT,X)(\epsilon-1)Pr(G=NT|X)\mathbb{E}[\tilde{Z}|X]\Big] \\
&\quad + \mathbb{E}\Big[\mathbb{E}\big[\big(\epsilon\eta_0'X+(1-\epsilon)\eta_1'X\big)\tilde{Z}\mid X\big]\Big] \\
&= \mathbb{E}\Big[\Delta(CP,X)Pr(G=CP|X)(\mathbb{E}[Z|X]+\epsilon-1)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(AT,X)\epsilon Pr(G=AT|X)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(NT,X)(\epsilon-1)Pr(G=NT|X)\mathbb{E}[\tilde{Z}|X]\Big] \\
&\quad + \epsilon\eta_0'\mathbb{E}[X\tilde{Z}]+(1-\epsilon)\eta_1'\mathbb{E}[X\tilde{Z}] \\
&= \mathbb{E}\Big[\Delta(CP,X)Pr(G=CP|X)(\mathbb{E}[Z|X]+\epsilon-1)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(AT,X)\epsilon Pr(G=AT|X)\mathbb{E}[\tilde{Z}|X] \\
&\quad + \Delta(NT,X)(\epsilon-1)Pr(G=NT|X)\mathbb{E}[\tilde{Z}|X]\Big].
\end{aligned}
$$

What remains are the treatment effect terms for the different compliance groups, each scaled by the projection error $\mathbb{E}[\tilde{Z}|X]$. This projection error therefore determines how the IV estimand places weight on the treatment effects of compliers, always-takers, and never-takers.

Step 6: Constructing the Numerator.

Recall the decomposition of the instrumental variables estimand numerator derived in Step 1:

$$\mathbb{E}[Y\tilde{Z}] = \mathbb{E}[Cov(Y,Z\mid X)] + \mathbb{E}\big[\mathbb{E}[Y\mid X]\mathbb{E}[\tilde{Z}\mid X]\big]. \tag{8}$$

From Step 2, the conditional covariance component evaluates to:

$$Cov(Y,Z\mid X) = \Delta(CP,X)\Pr(G=CP\mid X)\mathbb{E}[Z\mid X]\big(1-\mathbb{E}[Z\mid X]\big). \tag{9}$$

From Step 5, the expanded conditional mean component evaluates to:

$$\begin{aligned}\mathbb{E}\big[\mathbb{E}[Y \mid X]\mathbb{E}[\tilde{Z} \mid X]\big] = \mathbb{E}\Big[&\Delta(CP, X)\Pr(G = CP \mid X)\big(\mathbb{E}[Z \mid X] + \epsilon - 1\big)\mathbb{E}[\tilde{Z} \mid X] \\ &+ \Delta(AT, X)\epsilon \Pr(G = AT \mid X)\mathbb{E}[\tilde{Z} \mid X] \\ &+ \Delta(NT, X)(\epsilon - 1)\Pr(G = NT \mid X)\mathbb{E}[\tilde{Z} \mid X]\Big]. \end{aligned} \tag{10}$$

Substituting Equation 9 and Equation 10 into Equation 8 yields the complete expectation for the numerator:

$$\begin{aligned}\mathbb{E}[Y\tilde{Z}] = \mathbb{E}\Big[&\Delta(CP, X)\Pr(G = CP \mid X)\mathbb{E}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big) \\ &+ \Delta(CP, X)\Pr(G = CP \mid X)\big(\mathbb{E}[Z \mid X] + \epsilon - 1\big)\mathbb{E}[\tilde{Z} \mid X] \\ &+ \Delta(AT, X)\epsilon \Pr(G = AT \mid X)\mathbb{E}[\tilde{Z} \mid X] \\ &+ \Delta(NT, X)(\epsilon - 1)\Pr(G = NT \mid X)\mathbb{E}[\tilde{Z} \mid X]\Big].\end{aligned}$$

Factoring the common term $\Delta(CP, X)\Pr(G = CP \mid X)$ from the first two expressions isolates the complier coefficient:

$$\begin{aligned}\mathbb{E}[Y\tilde{Z}] = \mathbb{E}\Big[&\Delta(CP, X)\Pr(G = CP \mid X)\Big(\mathbb{E}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big) \\ &+ \big(\mathbb{E}[Z \mid X] + \epsilon - 1\big)\mathbb{E}[\tilde{Z} \mid X]\Big) \\ &+ \Delta(AT, X)\epsilon \Pr(G = AT \mid X)\mathbb{E}[\tilde{Z} \mid X] \\ &+ \Delta(NT, X)(\epsilon - 1)\Pr(G = NT \mid X)\mathbb{E}[\tilde{Z} \mid X]\Big].\end{aligned} \tag{11}$$

Step 7: Rewriting the Complier Coefficient.

Consider the bracketed coefficient multiplying the complier term in Equation 11:

$$\mathbb{E}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big) + \big(\mathbb{E}[Z \mid X] + \epsilon - 1\big)\mathbb{E}[\tilde{Z} \mid X]. \tag{12}$$

To reduce this expression, we first expand the second component by separating the ϵ term from the expected values:

$$\begin{aligned}\big(\mathbb{E}[Z \mid X] + \epsilon - 1\big)\mathbb{E}[\tilde{Z} \mid X] &= \big(\mathbb{E}[Z \mid X] - 1\big)\mathbb{E}[\tilde{Z} \mid X] + \epsilon\mathbb{E}[\tilde{Z} \mid X] \\ &= -\big(1 - \mathbb{E}[Z \mid X]\big)\mathbb{E}[\tilde{Z} \mid X] + \epsilon\mathbb{E}[\tilde{Z} \mid X].\end{aligned} \tag{13}$$

Substituting Equation 13 back into the original complier coefficient (Equation 12) allows

us to factor out the common term $\big(1 - \mathbb{E}[Z \mid X]\big)$:

$$\begin{aligned}
&\mathbb{E}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big) - \big(1 - \mathbb{E}[Z \mid X]\big)\mathbb{E}[\tilde{Z} \mid X] + \epsilon\mathbb{E}[\tilde{Z} \mid X] \\
&= \big(\mathbb{E}[Z \mid X] - \mathbb{E}[\tilde{Z} \mid X]\big)\big(1 - \mathbb{E}[Z \mid X]\big) + \epsilon\mathbb{E}[\tilde{Z} \mid X]. \qquad (14)
\end{aligned}$$

We now apply the definition of the instrument residual, $\tilde{Z} \equiv Z - \mathbb{L}[Z \mid X]$. Taking the conditional expectation of both sides yields the identity $\mathbb{E}[\tilde{Z} \mid X] = \mathbb{E}[Z \mid X] - \mathbb{L}[Z \mid X]$. Rearranging this identity isolates the linear projection:

$$\mathbb{L}[Z \mid X] = \mathbb{E}[Z \mid X] - \mathbb{E}[\tilde{Z} \mid X].$$

By substituting $\mathbb{L}[Z \mid X]$ into the first term of Equation 14, the complier coefficient simplifies to its final form:

$$\mathbb{L}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big) + \epsilon\mathbb{E}[\tilde{Z} \mid X].$$

Inserting this simplified coefficient back into Equation 11 provides the final expression for the numerator of the IV estimand, demonstrating that it is comprised strictly of weighted treatment effects for the three compliance groups:

$$\begin{aligned}
\mathbb{E}[Y\tilde{Z}] = \mathbb{E}\Big[&\Delta(CP, X)\Pr(G = CP \mid X)\Big(\epsilon\mathbb{E}[\tilde{Z} \mid X] + \mathbb{L}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big)\Big) \\
&+ \Delta(AT, X)\epsilon\Pr(G = AT \mid X)\mathbb{E}[\tilde{Z} \mid X] \\
&+ \Delta(NT, X)(\epsilon - 1)\Pr(G = NT \mid X)\mathbb{E}[\tilde{Z} \mid X]\Big]. \qquad (15)
\end{aligned}$$

Step 8: Dividing by the First-Stage Denominator.

Recall that the linear instrumental variables estimand, β_{IV}, is defined as the ratio of the reduced-form covariance to the first-stage covariance:

$$\beta_{IV} = \frac{\mathbb{E}[Y\tilde{Z}]}{\mathbb{E}[T\tilde{Z}]}.$$

To derive the final weighting scheme, we divide the expanded numerator from Equation 15 by the scalar constant $\mathbb{E}[T\tilde{Z}]$. Because the expectation operator is linear, we can

distribute the constant $\mathbb{E}[T\tilde{Z}]^{-1}$ across the three additive components of the expectation:

$$\begin{aligned}\beta_{IV} = \mathbb{E}&\left[\frac{\epsilon\mathbb{E}[\tilde{Z} \mid X] + \mathbb{L}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big)}{\mathbb{E}[T\tilde{Z}]} \Pr(G = CP \mid X)\Delta(CP, X)\right] \\ &+ \mathbb{E}\left[\frac{\epsilon\mathbb{E}[\tilde{Z} \mid X]}{\mathbb{E}[T\tilde{Z}]} \Pr(G = AT \mid X)\Delta(AT, X)\right] \\ &+ \mathbb{E}\left[\frac{(\epsilon - 1)\mathbb{E}[\tilde{Z} \mid X]}{\mathbb{E}[T\tilde{Z}]} \Pr(G = NT \mid X)\Delta(NT, X)\right]. \end{aligned} \tag{16}$$

By defining the weighting functions $\omega_\epsilon(G, X)$ as the exact multipliers attached to the conditional average treatment effects $\Delta(G, X)$ in Equation 16, we recover the precise formulation presented in the main text:

$$\beta_{IV} = \mathbb{E}\big[\omega_\epsilon(CP, X)\Delta(CP, X)\big] + \mathbb{E}\big[\omega_\epsilon(AT, X)\Delta(AT, X)\big] + \mathbb{E}\big[\omega_\epsilon(NT, X)\Delta(NT, X)\big],$$

where the weights are formally defined as:

$$\begin{aligned}&\omega_\epsilon(\mathrm{CP}, X) \equiv \Big(\epsilon\,\mathbb{E}[\tilde{Z} \mid X] + \mathbb{L}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big)\Big) \Pr(G = \mathrm{CP} \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1}, \\ &\omega_\epsilon(\mathrm{AT}, X) \equiv \epsilon\,\mathbb{E}[\tilde{Z} \mid X]\ \Pr(G = \mathrm{AT} \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1}, \\ &\omega_\epsilon(\mathrm{NT}, X) \equiv (\epsilon - 1)\,\mathbb{E}[\tilde{Z} \mid X]\ \Pr(G = \mathrm{NT} \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1}.\end{aligned}$$

This explicitly confirms that the TSLS estimand is an ϵ-indexed weighted average of the treatment effects for all compliance groups, completing the proof of Proposition 2. □

3.4 Proof of Proposition 1

The formulation of the instrumental variables estimand β_{IV} in Proposition 1 is a special case of the generalized decomposition in Proposition 2, obtained by evaluating the weights at $\epsilon = 1$.

Step 1: Never-Taker Weights. By substituting $\epsilon = 1$ into the generalized never-taker weight function from Proposition 2, the term evaluates to zero:

$$\begin{aligned}\omega_1(NT, X) &= (\epsilon - 1)\,\mathbb{E}[\tilde{Z} \mid X]\ \Pr(G = NT \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1} \\ &= (1 - 1)\,\mathbb{E}[\tilde{Z} \mid X]\ \Pr(G = NT \mid X)\,\mathbb{E}[\tilde{Z}T]^{-1} \\ &= 0.\end{aligned}$$

Consequently, the treatment effects of never-takers vanish from the decomposition.

Step 2: Complier Weights. Substituting $\epsilon = 1$ into the complier weight function

yields:

$$\omega_1(CP, X) = \Big(1 \cdot \mathbb{E}[\tilde{Z} \mid X] + \mathbb{L}[Z \mid X]\big(1 - \mathbb{E}[Z \mid X]\big)\Big) \Pr(G = CP \mid X)\, \mathbb{E}[\tilde{Z}T]^{-1}.$$

Applying the definition of the projection error, $\mathbb{E}[\tilde{Z} \mid X] = \mathbb{E}[Z \mid X] - \mathbb{L}[Z \mid X]$, the bracketed term simplifies as follows:

$$\begin{aligned}\mathbb{E}[Z \mid X] - \mathbb{L}[Z \mid X] + \mathbb{L}[Z \mid X] - \mathbb{L}[Z \mid X]\mathbb{E}[Z \mid X] &= \mathbb{E}[Z \mid X] - \mathbb{L}[Z \mid X]\mathbb{E}[Z \mid X] \\ &= \mathbb{E}[Z \mid X]\big(1 - \mathbb{L}[Z \mid X]\big).\end{aligned}$$

Thus, the complier weight is:

$$\omega_1(CP, X) = \mathbb{E}[Z \mid X]\big(1 - \mathbb{L}[Z \mid X]\big) \Pr(G = CP \mid X)\, \mathbb{E}[\tilde{Z}T]^{-1}.$$

To determine the sign of this weight, assume the first-stage covariance is positive ($\mathbb{E}[\tilde{Z}T] > 0$). The probability $\Pr(G = CP \mid X)$ is non-negative. Because the instrument Z is binary, its conditional expectation is bounded, such that $\mathbb{E}[Z \mid X] \in [0, 1]$. Therefore, the sign of $\omega_1(CP, X)$ depends entirely on the term $(1 - \mathbb{L}[Z \mid X])$. The complier weight is negative if and only if the linear projection exceeds one ($\mathbb{L}[Z \mid X] > 1$).

Step 3: Always-Taker Weights. Substituting $\epsilon = 1$ into the always-taker weight function yields:

$$\omega_1(AT, X) = \mathbb{E}[\tilde{Z} \mid X] \Pr(G = AT \mid X)\, \mathbb{E}[\tilde{Z}T]^{-1}.$$

Given $\Pr(G = AT \mid X) \geq 0$ and $\mathbb{E}[\tilde{Z}T] > 0$, the sign of the always-taker weight depends on the projection error $\mathbb{E}[\tilde{Z} \mid X]$.

By definition of ordinary least squares, because the covariate vector X contains a constant, the unconditional expectation of the projection error is zero:

$$\mathbb{E}\big[\mathbb{E}[\tilde{Z} \mid X]\big] = \mathbb{E}[\tilde{Z}] = 0. \tag{17}$$

For Equation 17 to hold, the projection error $\mathbb{E}[\tilde{Z} \mid X]$ must either equal zero with probability 1, or it must take both positive and negative values with positive probability. In the latter case, $\omega_1(AT, X)$ is negative for some values of X. Notably, this negativity condition holds regardless of whether the first-stage covariance $\mathbb{E}[\tilde{Z}T]$ is positive or negative.

Conclusion. Unless the linear projection equals the conditional expectation deterministically ($\mathbb{L}[Z \mid X] = \mathbb{E}[Z \mid X]$), the IV estimand incorporates negatively weighted treatment effects for always-takers and potentially compliers. □

3.5 Intuition Behind Propositions 1 and 2

Although some intuition has already appeared in the proofs, it is useful to summarize the main mechanisms more directly.

1. Projection Error versus Conditioning A nonparametric IV estimator flexibly conditions on the covariates X, which helps isolate the causal responses of compliers. TSLS, by contrast, relies on a linear projection $\mathbb{L}[Z|X]$. If this projection does not coincide with the conditional expectation $\mathbb{E}[Z|X]$, a projection error remains. Because TSLS relies on this linear projection, the resulting projection error propagates into the second stage and can introduce variation from non-compliers into the IV estimand.

2. Level Dependence As a consequence of this projection error, the TSLS estimand depends on the levels of the potential outcomes of non-compliers: the treated outcomes of always-takers $Y(1)$ and the untreated outcomes of never-takers $Y(0)$. Because the estimand becomes dependent on potential outcome levels rather than solely on treatment effect differences, it exhibits what is termed *level dependence*. Consequently, the TSLS estimand may produce a negative estimate even when all individual treatment effects are positive, particularly if the baseline outcomes of non-compliers are large relative to the complier effects.

3. Non-Unique Decompositions The level-dependent component contains outcomes from both always-takers and never-takers. Attempting to isolate pure treatment effects requires centering the conditional mean. Centering around the untreated baseline removes the never-takers, allocating the projection bias to always-takers (Proposition 1, $\epsilon = 1$). Conversely, centering around the treated baseline removes the always-takers, allocating the bias to never-takers ($\epsilon = 0$). Because both baselines are linear functions of X, both approaches provide mathematically valid representations of the TSLS estimand. A weighted average of these centerings yields the ϵ-indexed continuum of decompositions established in Proposition 2, each corresponding to a different allocation of the projection-error component across compliance groups.

In summary, when the rich covariates condition fails, the projection error induces level dependence in the TSLS estimand, eliminates a unique decomposition into treatment effects, and introduces the possibility of negative weights.

3.6 Weakly Causal Definition

As demonstrated, standard TSLS specifications generally fail to identify a non-negatively weighted average of treatment effects. Furthermore, if the rich covariates condition fails or the assumption of linear potential outcomes is relaxed, the TSLS estimand admits

non-unique or nonexistent decompositions. This structural instability necessitates a minimal criterion for weak causality to determine whether an estimand retains any valid interpretive value.

To formalize this requirement, Blandhol et al. (2022) introduce the concept of group treatment responses (GTRs). Let $\mu_j(g, x) \equiv \mathbb{E}[Y(t_j) \mid G = g, X = x]$ denote the expected potential outcome for treatment level t_j, compliance group g, and covariate vector x. A minimally valid causal estimand must be directionally consistent with these underlying responses.

Definition WC (Weakly Causal): An estimand β is weakly causal if, for any set of group treatment responses μ, both of the following conditions hold (Blandhol et al., 2022):

1. If $\mu_j(g, x) - \mu_{j-1}(g, x) \geq 0$ for all $j \geq 1$, $g \in \mathcal{G}$, and $x \in \mathcal{X}$, then $\beta \geq 0$.
2. If $\mu_j(g, x) - \mu_{j-1}(g, x) \leq 0$ for all $j \geq 1$, $g \in \mathcal{G}$, and $x \in \mathcal{X}$, then $\beta \leq 0$.

In intuitive terms, Definition WC requires that an estimand must not be systematically misleading about the sign of the underlying treatment effects. If a treatment has a universally positive or zero effect on every individual in every subgroup, a weakly causal estimand must yield a non-negative value.

3.7 Theorem 1

Building on the fragility demonstrated in Propositions 1 and 2, Blandhol et al. (2022) establish their main theoretical result, which formally links the weakly causal requirement to the rich covariates condition. First, we formalize the core identifying assumptions of the instrumental variables framework.

Let $G \equiv (T(z_0), T(z_1), \ldots, T(z_K))$ denote an individual's compliance group, representing their configuration of potential treatment choices across all possible instrument realizations.

Assumption EXO (Exogeneity). Conditional on covariates X, the instrument Z is jointly independent of the compliance group and all potential outcomes (Blandhol et al., 2022):

$$(G, \{Y(t)\}_{t \in \mathcal{T}}) \perp Z \mid X.$$

Assumption MON (Monotonicity). Ordering the support of the instrument such that $z_0 \leq z_1 \leq \cdots \leq z_K$, an increase in the instrument weakly increases treatment for all individuals, conditional on X (Blandhol et al., 2022):

$$\mathbb{P}\big[T(z_0) \leq T(z_1) \leq \cdots \leq T(z_K) \mid X = x\big] = 1 \quad \text{for all } x.$$

The main result of Blandhol et al. (2022) is established in Theorem 1.

Theorem 1 (Blandhol et al., 2022). Suppose that Assumptions EXO and MON are satisfied. Then the linear IV estimand β_{IV} is weakly causal *if and only if* the IV specification has rich covariates.

Theorem 1 establishes that the rich covariates condition is both necessary and sufficient for the linear IV estimand to maintain a weakly causal interpretation. This condition holds in only two specific cases (Kolesár, 2013). The first is when the covariate vector X constitutes a fully saturated specification of indicator variables. The second is when the instrument Z is mean-independent of the covariates ($\mathbb{E}[Z \mid X = x] = \mathbb{E}[Z]$). Because X contains a constant, the linear projection in this second case simplifies exactly to the conditional mean:

$$\mathbb{L}[Z \mid X = x] \equiv \mathbb{E}[ZX']\mathbb{E}[XX']^{-1}x = \mathbb{E}[Z]\mathbb{E}[1X']\mathbb{E}[XX']^{-1}x = \mathbb{E}[Z].$$

Outside these two scenarios, the assumption of rich covariates is an implicit parametric assumption that requires empirical justification.

3.8 Proof of Theorem 1

To prove Theorem 1, the authors introduce two intermediate propositions that characterize the exact conditions under which any estimand can be considered weakly causal.

First, **Proposition 3** states that under the EXO Assumption, any estimand of the form $\beta = \mathbb{E}[b(T, X, Z)Y]$ can be decomposed into a baseline level term and a weighted sum of treatment contrasts (Blandhol et al., 2022):

$$\beta = \sum_{g,x} \omega_0(g,x)\mu_0(g,x) + \sum_{g,x}\sum_{j=1}^{J} \omega_j(g,x)\big(\mu_j(g,x) - \mu_{j-1}(g,x)\big) \tag{18}$$

For the specific case of the linear IV estimand (β_{IV}), the weights in this decomposition are given by:

$$\omega_j(g,x) = \mathbb{E}[\tilde{Z}T]^{-1}\mathbb{E}\big[1[T \geq t_j]\tilde{Z} \mid G = g, X = x\big]Pr(G = g, X = x) \tag{19}$$

Next, **Proposition 4** establishes under the EXO Assumption that an estimand of the form in Equation 18 is weakly causal if and only if two conditions hold simultaneously (Blandhol et al., 2022):

1. **Level Independence:** The weight on the baseline levels must be exactly zero, meaning $\omega_0(g,x) = 0$ for all g, x.

2. **Non-Negative Weights:** The weights on the treatment contrasts must be strictly non-negative, meaning $\omega_j(g, x) \geq 0$ for all $j \geq 1, g, x$.

The proof of Theorem 1 proceeds by evaluating these two requirements using the specific weight formula defined in Equation 19.

Step 1: Necessity (Level Independence implies Rich Covariates)

For β_{IV} to be weakly causal, Proposition 4 dictates that it must satisfy level independence: $\omega_0(g, x) = 0$ for all compliance groups g and covariates x.

We start by evaluating the weight formula (Equation 19) at $j = 0$. Since treatment levels are bounded such that $T \geq t_0$ is always true, the indicator function $1[T \geq t_0] = 1$. Substituting this into the weight equation yields:

$$\omega_0(g, x) = \mathbb{E}[\tilde{Z}T]^{-1}\mathbb{E}[\tilde{Z} \mid G = g, X = x]Pr(G = g \mid X = x) = 0 \tag{20}$$

For this entire expression to equal zero, the inner conditional expectation must be zero, because the first-stage denominator $\mathbb{E}[\tilde{Z}T]^{-1}$ is non-zero, and for any x, there exists at least one group g where $Pr(G = g \mid X = x) > 0$. Thus, level independence simplifies to the requirement:

$$\mathbb{E}[\tilde{Z} \mid G = g, X = x] = 0 \tag{21}$$

Now, we apply the EXO assumption, which states that the compliance groups G are independent of the instrument Z conditional on X. Because $\tilde{Z} \equiv Z - \mathbb{L}[Z|X]$ is purely a function of Z and X, $\tilde{Z}$ is also independent of G given X. This allows us to drop the conditioning on G:

$$\mathbb{E}[\tilde{Z} \mid G = g, X = x] = \mathbb{E}[\tilde{Z} \mid X = x] \tag{22}$$

By substituting the definition of the instrument residual, $\tilde{Z} = Z - \mathbb{L}[Z|X]$, back into Equation 22, we obtain:

$$\begin{aligned}
\mathbb{E}[\tilde{Z} \mid X = x] &= 0 \\
\mathbb{E}\big[Z - \mathbb{L}[Z|X] \mid X = x\big] &= 0 \\
\mathbb{E}[Z \mid X = x] - \mathbb{L}[Z \mid X = x] &= 0 \\
\mathbb{E}[Z \mid X = x] &= \mathbb{L}[Z \mid X = x]
\end{aligned} \tag{23}$$

Equation 23 perfectly recovers the definition of rich covariates. Because the compliance groups exhaustively partition the population, if covariates are not rich, $\omega_0(g, x)$ will inevitably be non-zero for some group, meaning β_{IV} suffers from level dependence and fails to be weakly causal. Thus, rich covariates are a strictly necessary condition.

Step 2: Sufficiency (Rich Covariates implies Non-Negative Weights)

To establish sufficiency, we assume the rich covariates condition holds ($\mathbb{E}[Z|X] = \mathbb{L}[Z|X]$) and must prove that the non-negative weights condition from Proposition 4 is satisfied ($\omega_j(g,x) \geq 0$ for all j, g, x).

Let $\mathcal{Z}_j(g)$ denote the specific set of instrument values that induce an individual in compliance group g to select treatment level t_j or higher. We can replace the treatment indicator $1[T \geq t_j]$ with the instrument set indicator $1[Z \in \mathcal{Z}_j(g)]$. Substituting this into our weight equation:

$$\omega_j(g,x) = \mathbb{E}[\tilde{Z}T]^{-1}\mathbb{E}\big[\tilde{Z}1[Z \in \mathcal{Z}_j(g)] \mid G = g, X = x\big]Pr(G = g, X = x) \tag{24}$$

Using the EXO assumption, we drop the conditioning on G inside the expectation. Furthermore, because we assume rich covariates hold, we know that $\mathbb{E}[\tilde{Z} \mid X = x] = 0$. This implies that the conditional expectation of the product can be perfectly rewritten as a conditional covariance:

$$\mathbb{E}\big[\tilde{Z}1[Z \in \mathcal{Z}_j(g)] \mid X = x\big] = Cov\big(Z, 1[Z \in \mathcal{Z}_j(g)] \mid X = x\big) \tag{25}$$

Plugging Equation 25 back into Equation 24, the weight simplifies to:

$$\omega_j(g,x) \propto \mathbb{E}[\tilde{Z}T]^{-1}Cov\big(Z, 1[Z \in \mathcal{Z}_j(g)] \mid X = x\big) \tag{26}$$

To prove $\omega_j(g,x) \geq 0$, we must evaluate the signs of both the denominator and the numerator in Equation 26.

Sign of the Denominator: The term $\mathbb{E}[\tilde{Z}T]$ can be expanded using the law of iterated expectations and the definition of $\tilde{Z}$ under rich covariates:

$$\begin{aligned}\mathbb{E}[\tilde{Z}T] &= \mathbb{E}\big[(Z - \mathbb{E}[Z|X])\mathbb{E}[T|X,Z]\big] \\ &= \mathbb{E}\big[Cov(Z, \mathbb{E}[T|X,Z] \mid X)\big]\end{aligned} \tag{27}$$

The MON assumption dictates that higher values of the instrument never decrease the chosen treatment level. Therefore, $\mathbb{E}[T|X,Z]$ must be a weakly increasing function of Z. Because the covariance of two weakly increasing functions is always non-negative, Equation 27 guarantees that $\mathbb{E}[\tilde{Z}T] \geq 0$.

Sign of the Numerator: We must determine the sign of the covariance term $Cov\big(Z, 1[Z \in \mathcal{Z}_j(g)] \mid X = x\big)$. Suppose $Pr(G = g \mid X = x) > 0$. We must prove that the indicator function $z \mapsto 1[z \in \mathcal{Z}_j(g)]$ is weakly increasing in z. We can prove this by contradiction. Suppose the function is *not* weakly increasing. Then there must exist

two instrument values, $z < z'$, such that:

$$z \in \mathcal{Z}_j(g) \quad \text{and} \quad z' \notin \mathcal{Z}_j(g) \tag{28}$$

By the definition of the set $\mathcal{Z}_j(g)$, Equation 28 strictly implies that the treatment choice at the lower instrument value is higher than the treatment choice at the higher instrument value:

$$T(z) \geq t_j > T(z')$$

Given $\Pr(G = g \mid X = x) > 0$, this implies

$$\Pr\big(T(z) \geq t_j > T(z') \mid X = x\big) \geq \Pr(G = g \mid X = x) > 0,$$

which violates the MON assumption, which requires $T(z') \geq T(z)$ when $z' > z$.

Therefore, our premise was false, and the indicator function $1[Z \in \mathcal{Z}_j(g)]$ must be weakly increasing in Z.

Because both Z and the indicator function are weakly increasing, their conditional covariance is non-negative:

$$Cov\big(Z, 1[Z \in \mathcal{Z}_j(g)] \mid X = x\big) \geq 0$$

Conclusion: Since both the denominator ($\mathbb{E}[\tilde{Z}T] \geq 0$) and the numerator (the conditional covariance ≥ 0) are non-negative, the resulting weight must be non-negative:

$$\omega_j(g, x) \geq 0 \quad \text{for all } j, g, x$$

By Proposition 4, satisfying both level independence and non-negative weights guarantees that β_{IV} is weakly causal. This concludes the proof, cementing the core takeaway: without rich covariates, the linear IV estimand is mathematically compromised.

4 Critical Discussion and Practical Limitations

Theorem 1 establishes a mathematical boundary for the causal interpretation of the linear IV estimand, but its practical implications present challenges for empirical research. This section examines the limitations of the proposed solutions and their implications for empirical practice.

4.1 The Minimal Nature of the Weakly Causal Criterion

The weakly causal requirement is a minimal baseline. An estimand can satisfy this condition and lack empirical relevance; for example, the trivial estimand $\beta = 0$ is weakly

causal. However, an estimand that fails this criterion cannot be interpreted as reflecting the causal effect of T on Y, as it lacks mathematical guarantee of directional consistency. Failing a threshold this permissive underscores the severity of the structural vulnerabilities in non-saturated specifications.

4.2 The Empirical Infeasibility of Saturated Specifications

According to Kolesár (2013), the rich covariates condition holds mechanically in a saturated covariate specification. The most robust implementation is the "saturate and weight" approach (Angrist and Pischke, 2009), which saturates the first stage in both covariates and instrument-covariate interactions. This interaction is mathematically necessary to preserve causal interpretations under weaker forms of monotonicity where the direction of the instrument's effect varies across covariate strata (Słoczyński, 2020).

However, this approach is severely limited by the curse of dimensionality. Including continuous covariates or numerous discrete controls causes the number of parameter strata to grow exponentially. Furthermore, saturating the first-stage regression with instrument-covariate interactions generates a high-dimensional instrument vector. As the number of excluded instruments grows relative to the sample size, TSLS estimates suffer from many-instruments bias, reverting toward OLS estimates and undermining the identification strategy.

4.3 The Restrictiveness of Mean Independence

The second case where the rich covariates condition holds is when the instrument Z is mean-independent of the covariates: $\mathbb{E}[Z \mid X = x] = \mathbb{E}[Z]$ (Kolesár, 2013). Because the covariate vector X contains a constant, the linear projection in this scenario simplifies exactly to the conditional mean:

$$\mathbb{L}[Z \mid X = x] \equiv \mathbb{E}[ZX']\mathbb{E}[XX']^{-1}x = \mathbb{E}[Z]\mathbb{E}[1X']\mathbb{E}[XX']^{-1}x = \mathbb{E}[Z].$$

While mathematically sound, this condition is restrictive in applied empirical work. In observational research designs, covariates are typically included precisely because the instrument is not unconditionally independent of potential confounders. If Z were already mean-independent of X, conditioning on X would be largely redundant for identification purposes. Consequently, in settings where covariates are necessary to satisfy the exogeneity assumption, this exception is generally inapplicable.

4.4 The Fragility of Parametric Restrictions

Outside these two specific scenarios—saturated models and mean independence—the assumption of rich covariates reduces to an implicit parametric assumption requiring empirical justification (Blandhol et al., 2022). Specifically, it requires that the true conditional expectation of the instrument, $\mathbb{E}[Z|X]$, is linear in X.

This reliance on functional form is problematic. Economic theory rarely dictates the functional relationship between the instrument and covariates. Assuming the linear projection matches the conditional mean is therefore vulnerable to misspecification. If this parametric restriction fails, Propositions 1 and 2 establish that the TSLS estimand becomes level-dependent and assigns negative weights to treatment effects, violating the weakly causal requirement.

Diagnostic Recommendations. When a flexible, saturated covariate specification cannot be utilized due to dimensionality constraints, researchers must rigorously defend the implicit parametric assumption that $\mathbb{L}[Z|X] = \mathbb{E}[Z|X]$. To detect violations of this condition, Blandhol et al. (2022) recommend two routine diagnostic checks. First, for binary instruments, researchers must verify that all first-stage fitted values lie strictly within the logical $[0, 1]$ interval; values outside this range mechanically induce negative complier weights. Second, researchers should formally evaluate the null hypothesis of correct functional form using Ramsey's RESET test. Despite its computational simplicity, the survey of empirical practice indicates that these basic specification tests are systematically omitted from applied IV research.

4.5 Alternative Methodological Approaches

To address the limitations of the linear IV estimator when the rich covariates condition fails, or to recover more interpretable parameters when it holds, the literature proposes several alternative estimators.

Partially Linear IV (PLIV) and Machine Learning. If the rich covariates condition is not satisfied, researchers can target β_{rich}, which represents the IV estimand under the theoretical satisfaction of the rich covariates condition. While estimating β_{rich} via a fully saturated covariate specification (Angrist, 1998; Angrist and Pischke, 2009) is statistically valid, it introduces a severe bias-variance trade-off in high-dimensional settings. To overcome this, Chernozhukov et al. (2018) propose the partially linear IV (PLIV) model. By utilizing Neyman orthogonal scores and cross-fitting, the PLIV framework allows researchers to estimate the nuisance functions (the conditional expectations of the outcome, treatment, and instrument) nonparametrically using supervised machine learning algorithms. The resulting coefficient on treatment yields a consistent estimate of β_{rich}.

Unconditional Average Causal Response (ACR). Even when the rich covariates condition holds, β_{rich} weights covariate-specific treatment effects proportionally to the conditional variance of the instrument, $V[Z|X]$. Because this variance-driven weighting lacks a clear counterfactual interpretation and can yield misleading estimates (Słoczyński, 2020, 2024), researchers may instead target the ACR (Angrist and Imbens, 1995). When treatment is binary, the ACR corresponds to the LATE.

The ACR can be estimated using two primary methods based on the nonparametric identification results of Tan (2006) and Frölich (2007). The first utilizes Double/Debiased Machine Learning (DDML) to estimate the necessary orthogonal scores nonparametrically (Chernozhukov et al., 2018). The second employs a weight-normalized inverse propensity score estimator, which requires parameterizing the instrument propensity score, $\mathbb{E}[Z|X]$ (Uysal, 2015; Heller, 2022; Słoczyński, 2024).

Abadie's κ-Weighting. For settings with binary treatment and binary instruments, Abadie (2003) and Angrist and Pischke (2009) propose a weighted linear regression to recover complier-specific effects. This approach weights a regression of the outcome on the treatment and covariates by κ, defined as:

$$\kappa = 1 - \frac{T(1-Z)}{1-\mathbb{E}[Z|X]} - \frac{(1-T)Z}{\mathbb{E}[Z|X]}.$$

In the population, this κ-weighted regression is equivalent to an unweighted regression on the subpopulation of compliers. However, Blandhol et al. (2022) demonstrate that the mathematical necessity of the rich covariates condition extends to this framework; rich covariates must hold for the κ-weighted estimand to remain weakly causal.

4.6 The Disconnect Between Theory and Empirical Practice

To contextualize the practical relevance of Theorem 1, Blandhol et al. (2022) surveyed 112 empirical papers utilizing TSLS published in five leading economics journals between 2000 and 2018. The survey reveals a severe disconnect between theoretical requirements and applied practice. While nearly all papers condition on covariates, and approximately one-third explicitly claim a LATE interpretation, only 5% of the surveyed papers employ a covariate-saturated specification. Furthermore, only a single paper utilizes a fully saturated first-stage. This demonstrates that the strict structural conditions required to avoid non-causal contamination are routinely violated in standard empirical work.

5 Empirical Application: Re-evaluating Krueger (1999)

This section presents the results of the replication of Krueger (1999) and evaluates the causal validity of the findings through the framework of Blandhol et al. (2022). The

analysis compares standard OLS and TSLS estimates with more robust estimators to determine the extent of specification bias induced by parametric covariate inclusion.

5.1 Data Source and Sample Characteristics

The empirical analysis utilizes data from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment Krueger (1999). The study randomly assigned students and teachers within schools to three distinct class types: small classes (13–17 students), regular classes (22–25 students), and regular classes with a full-time teacher aide. The longitudinal study commenced with a kindergarten cohort in the 1985–1986 school year and admitted entering students in subsequent grades Krueger (1999).

The primary outcome variable is the percentile rank of the average Stanford Achievement Test (SAT) score, computed from reading, mathematics, and word recognition components. Treatment is defined by indicators for small class assignment and regular class with an aide, with regular classes serving as the reference category. Control variables consist of student characteristics (White/Asian, female, free-lunch status, and age) and teacher attributes (race, experience, and degree attainment). To account for the within-school randomization design, all specifications incorporate school fixed effects.

Table 1 summarizes the realized class sizes and associated test score improvements. Realized small classes average approximately 15–16 students, while regular classes average 22–24. The raw achievement gain associated with small classes is positive across all grades and peaks in the first grade. Descriptive statistics for the full estimation sample, provided in Table 2, show an average SAT percentile score of approximately 51 and a treatment share of roughly 30% across grade levels.

Table 1: Class Size and Score Improvement by Grade

Grade	Small class	Regular class	Δ Score
K	15.1	22.3	+5.4
1	15.9	22.6	+7.0
2	15.4	23.4	+3.4
3	16.0	24.0	+2.1

Notes: Table reports average realized class size and raw test score improvements in percentile points. Values are computed from the Tennessee STAR data following the specifications in Krueger (1999).

5.2 Econometric Framework

The baseline analysis utilizes an education production function to estimate the impact of class size on achievement. Following Krueger (1999), the OLS specification is defined as

$$Y_{isg} = \beta_0 + \beta_1 Small_{isg} + \beta_2 RegAide_{isg} + X_{isg}\gamma + \alpha_s + \varepsilon_{isg}, \tag{29}$$

where Y_{isg} is the percentile score of student i in school s and grade g, and X_{isg} represents the vector of student and teacher covariates. The parameter β_1 identifies the causal effect of small classes under the assumption of perfect compliance with the experimental design.

Non-random transitions between class types and enrollment fluctuations introduce potential endogeneity in realized class size (Krueger, 1999). To address this, an IV approach utilizes initial random assignment as an instrument for realized class type. The first-stage regression is

$$T_{isg} = \pi_0 + \pi_1 Z_{isg} + X_{isg}\delta + \eta_s + u_{isg}, \tag{30}$$

where T_{isg} is the realized treatment and Z_{isg} represents the initial assignment indicators. The second-stage regression uses the predicted values from the first stage, $\widehat{T}_{isg}$, to identify the causal effect:

$$Y_{isg} = \beta_0 + \beta_1 \widehat{T}_{isg} + X_{isg}\gamma + \alpha_s + \varepsilon_{isg}. \tag{31}$$

The resulting TSLS estimand corrects for bias from non-compliance, provided the instrument satisfies the exclusion restriction and exogeneity conditions.

Crucially, while Krueger (1999) interprets these TSLS coefficients as LATE, the validity of this interpretation in the presence of parametric controls X_{isg} depends on the rich covariates condition explored by Blandhol et al. (2022). This section evaluates the robustness of the STAR results by testing this condition and implementing the non-parametric adjustments detailed in the following section.

5.3 Standard OLS and TSLS Replication

Table 2 displays the results of the baseline specifications across kindergarten through third grade. The OLS estimates in the first row, which include the full set of student and teacher controls alongside school fixed effects, show a persistent and statistically significant achievement gain for students in small classes. The effect size is most pronounced in first grade, at 7.38 percentile points, while remaining approximately 5.5 to 5.8 points in other grades.

The second and third rows report the TSLS estimates. When covariates and school fixed effects are omitted (Row 2), the IV estimates for grades 1–3 are substantially higher than the OLS counterparts, reaching 8.1 points in first grade. However, the standard errors increase significantly, reflecting the reduced precision of the unconditional estimator. In Row 3, which replicates the primary "IV with covariates" specification from Krueger

(1999), the estimates align closely with the OLS results. In kindergarten, the IV estimate of 5.54 is identical to the OLS result, while in third grade, the estimate of 7.30 suggests that the achievement gain is robust even when accounting for non-random switching across class types.

Table 2: Comparison of IV estimates for Project STAR (Krueger 1999 replication)

	(1)	(2)	(3)	(4)
	Kindergarten	Grade 1	Grade 2	Grade 3
OLS	5.540 (0.710)	7.380 (0.680)	5.530 (0.700)	5.850 (0.720)
IV, no covariates	5.060 (0.830)	8.100 (0.910)	6.920 (0.960)	7.900 (1.010)
IV, with covariates	5.540 (0.720)	7.100 (0.790)	6.370 (0.850)	7.300 (0.940)
PLIV (DDML)	5.214 (0.819)	5.809 (0.749)	4.624 (0.753)	4.918 (0.771)
LATE/ACR (DDML)	4.989 (0.729)	7.282 (0.795)	6.641 (0.855)	6.722 (0.933)
LATE/ACR (IPSW)	5.529	7.234	6.655	7.459
Ramsey RESET test p-val.				
(H_0 : rich covariates)	0.000	0.000	0.002	0.000
Relative specification bias	0.059	0.182	0.274	0.326
Specification vs. selection bias	1.000	0.822	1.927	2.556
Outcome variable		Average SAT Percentile Score		
Outcome variable, mean	51.400	51.500	51.200	51.000
Treatment variable		Small Class Indicator		
Treatment variable, mean	0.302	0.281	0.297	0.326
Included variables	84	81	79	80
Sample size	5,859	6,622	6,273	6,290

Notes: Standard errors are shown in parentheses. "IV, no covariates" models do not include school fixed effects or demographic controls. "IV, with covariates" and OLS models include school fixed effects, student, and teacher characteristics. Relative specification bias is calculated as $|\beta_{iv} - \beta_{rich}|/|\beta_{iv}|$. Specification vs. selection bias is calculated as $|\beta_{iv} - \beta_{rich}|/|\beta_{ols} - \beta_{rich}|$. Included variables represent the sum of school fixed effects and standard student/teacher controls. IPSW standard errors are omitted as analytical computation requires specific bootstrapping procedures outside the current scope.

5.4 Testing the Rich Covariates Condition

The central critique of Blandhol et al. (2022) is that the TSLS estimand is only weakly causal if the rich covariates condition holds, requiring the linear projection of the instrument to equal its conditional expectation. To test this, we apply the Ramsey RESET test to the linear first-stage regression of the instrument on covariates.

The results, reported in the middle panel of Table 2, provide evidence against the rich covariates assumption. The p-values are 0.000 for kindergarten, first grade, and third grade, and 0.002 for second grade. These results provide strong evidence against the functional form implied by the linear first-stage specification. Following Blandhol et al. (2022), rejection of the Ramsey RESET test indicates that the linear projection $\mathbb{L}[Z|X]$

is unlikely to coincide with the conditional expectation $\mathbb{E}[Z|X]$, suggesting that the rich covariates condition may not hold. Consequently, the linear TSLS specifications used in Krueger (1999) do not satisfy the sufficient conditions for TSLS to admit a standard LATE interpretation, and the resulting estimand may fail to admit the standard non-negatively weighted LATE interpretation.

5.5 Alternative Causal Estimands: DDML and IPSW

Then, we utilize the partially linear IV model estimated via Double/Debiased Machine Learning, as suggested by Blandhol et al. (2022). This approach flexibly estimates the conditional expectations $\mathbb{E}[Y|X]$ and $\mathbb{E}[Z|X]$ using machine learning methods, thereby relaxing the restrictive linear functional form imposed by standard TSLS.

The DDML estimates for β_{rich} in Row 4 are systematically lower than the standard TSLS estimates. In first grade, the estimate drops from 7.10 to 5.81, while in second grade, the effect falls to 4.62. This downward shift is consistent with the presence of specification bias induced by the parametric first-stage approximation. The LATE/ACR (DDML) estimates in Row 5, which target the average causal response without the variance-driven weighting of TSLS, show a similar pattern, with estimates ranging from 4.99 in kindergarten to 7.28 in first grade. The IPSW results in Row 6 further corroborate these magnitudes, particularly in the later grades.

5.6 Quantifying Specification Bias

The final panel of Table 2 quantifies the magnitude of the parametric misspecification. The relative specification bias—defined as

$$\frac{|\beta_{iv} - \beta_{rich}|}{|\beta_{iv}|}$$

—increases monotonically across grades, rising from 0.059 in kindergarten to 0.326 in third grade. This indicates that by the end of the experiment, nearly one-third of the estimated treatment effect is attributable to specification error rather than causal variation.

As shown in Figure 1, these magnitudes align with the cross-study evidence documented by Blandhol et al. (2022). While the kindergarten estimate is comparable to applications with minimal bias (e.g., Card, 1995), the later grades exhibit substantial discrepancies similar to those documented in Bloom et al., 2013 and Dal Bó et al., 2009. These results suggest that the specification bias observed in the STAR replication reflects a broader empirical pattern in the instrumental variables literature.

To benchmark the economic significance of this bias, we compare specification bias to

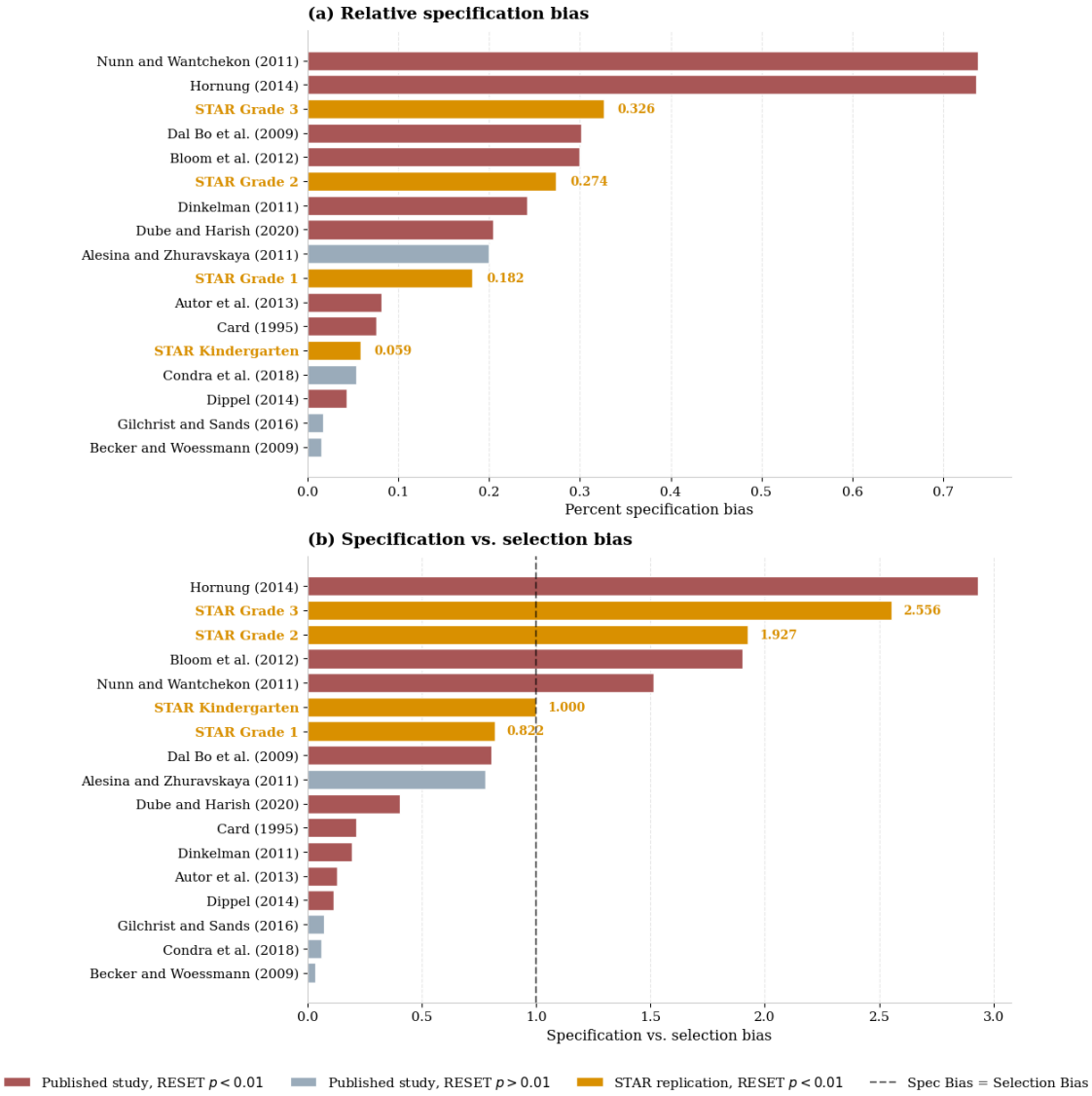

Figure 1: Relative magnitude of specification bias across applications

Notes: Published-study values are computed from Table SA.3 in Blandhol et al. (2022). STAR values are computed from the Project STAR replication table in this paper. Panel (a) plots $|\beta_{iv} - \beta_{rich}|/|\beta_{iv}|$. Panel (b) plots $|\beta_{iv} - \beta_{rich}|/|\beta_{ols} - \beta_{rich}|$. Bars are shaded according to whether the Ramsey RESET test rejects the null hypothesis of weak causality at the 1% level. The dashed vertical line at 1 in Panel (b) marks the threshold at which specification bias equals selection bias.

the estimated selection bias,

$$\frac{|\beta_{iv} - \beta_{rich}|}{|\beta_{ols} - \beta_{rich}|}.$$

In the second and third grades, this ratio reaches 1.927 and 2.556, respectively. These values imply that the distortion introduced by the linear covariate specification is more than twice as large as the endogeneity bias the IV strategy was intended to address. Such high ratios are uncommon in the survey conducted by Blandhol et al. (2022) and appear only in a few cases, most notably in Hornung, 2014.

The failure of the RESET diagnostics in all grade cohorts reinforces these findings. As in other applications exhibiting large specification bias, the rejection of the rich covariates condition in the STAR data ($p < 0.01$) indicates that the TSLS estimand cannot be interpreted as weakly causal. Collectively, these results demonstrate that in high-dimensional settings with many controls or fixed effects, misspecification of the covariate structure can be as consequential as the endogeneity problem itself, potentially leading to an inflated interpretation of the causal effect.

6 Conclusion

This seminar paper has interrogated the mathematical and empirical foundations of the LATE interpretation in the presence of covariates. While the benchmark theory of Imbens and Angrist (1994) provides a clean causal identification result for the unconditioned Wald estimator, the transition to standard TSLS with linear controls is fraught with structural vulnerabilities. As demonstrated by Blandhol et al. (2022), the standard TSLS estimand is only weakly causal if the rich covariates condition holds—a requirement that is systematically violated in most empirical applications.

The replication of Krueger (1999) provides a compelling case study of these theoretical concerns. Although the randomized nature of Project STAR is often cited as a gold standard for causal inference, the inclusion of linear school fixed effects and demographic controls fails the Ramsey RESET test in every grade cohort. This failure implies that the resulting TSLS estimates do not identify a non-negatively weighted average of treatment effects, but are instead contaminated by the baseline outcome levels of always-takers and never-takers. The subsequent re-estimation via DDML reveals that standard TSLS estimates are up to 32% larger than the rich-covariate benchmark in later grades. Crucially, the discovery that specification bias exceeds selection bias in the second and third grade samples underscores that the choice of functional form is as vital to causal validity as the choice of the identification strategy itself.

The results of Blandhol et al. (2022) have several implications for applied instrumental variables research.

1. **Careful treatment of covariates.** Because the causal interpretation of linear

IV with covariates depends on the rich covariates condition, researchers should think carefully about why particular covariates are included in the specification. When covariates are not essential for establishing instrument exogeneity, reporting estimates without them can provide a useful benchmark. When covariates are necessary for identification, their inclusion should be theoretically motivated rather than purely mechanical.

2. **Evaluation of first-stage specification.** Since the key requirement for causal interpretation concerns the relationship between the instrument and the covariates, it is useful to examine whether the linear projection used in TSLS adequately approximates the conditional expectation of the instrument. Diagnostic tools such as the Ramsey RESET test can provide evidence on potential misspecification.

3. **Use of more flexible estimators when necessary.** If the linear first-stage specification appears implausible, semi-parametric approaches can be used to estimate causal parameters that rely less strongly on functional form assumptions. Methods such as Double/Debiased Machine Learning allow covariates to be partialled out using flexible machine learning techniques.

4. **Focus on interpretable causal parameters.** In settings with binary instruments, researchers can also estimate unconditional causal parameters, such as the Average Causal Response. These parameters coincide with the unconditional LATE when the treatment is binary and avoid the variance-weighted interpretation associated with the conventional TSLS estimand.

Together, these considerations emphasize that the interpretation of IV estimates with covariates depends critically on how the relationship between the instrument and the covariates is specified.

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A Appendix

A.1 Proof of Proposition 3

Note that treatment T is stochastic only through the instrument Z once the compliance group G and covariates X are fixed, which follows directly from the definition of compliance types. Assumption EXO therefore implies that T and the potential outcomes $Y(t)$ are independent conditional on (G, X).

We begin by expressing the estimand as

$$\beta = \mathbb{E}[b(T, X, Z)Y]. \tag{32}$$

Applying the law of iterated expectations and conditioning on (G, X) yields

$$\beta = \sum_{g,x} \mathbb{E}[b(T, x, Z)Y \mid G = g, X = x] \Pr(G = g, X = x). \tag{33}$$

Because the realized outcome equals the potential outcome corresponding to the realized treatment level, we can write

$$Y = Y(T) = \sum_{j} 1[T = t_j]Y(t_j). \tag{34}$$

Substituting this expression into the previous equation gives

$$\beta = \sum_{g,x} \mathbb{E}\left[b(T,x,Z) \sum_j 1[T=t_j] Y(t_j) \mid G=g, X=x \right] \Pr(G=g, X=x) \tag{35}$$

$$= \sum_{g,x,j} \mathbb{E}[1[T=t_j] b(t_j,x,Z) Y(t_j) \mid G=g, X=x] \Pr(G=g, X=x). \tag{36}$$

Using Assumption EXO, the potential outcome $Y(t_j)$ is independent of (T,Z) conditional on (G,X), which allows the expectation to factorize:

$$\beta = \sum_{g,x,j} \mu_j(g,x) \mathbb{E}[1[T=t_j] b(t_j,x,Z) \mid G=g, X=x] \Pr(G=g, X=x), \tag{37}$$

where

$$\mu_j(g,x) \equiv \mathbb{E}[Y(t_j) \mid G=g, X=x]$$

denotes the group treatment response.

Define the weights

$$\psi_j(g,x) \equiv \mathbb{E}[1[T=t_j] b(t_j,x,Z) \mid G=g, X=x] \Pr(G=g, X=x). \tag{38}$$

Substituting this definition yields

$$\beta = \sum_{g,x,j} \mu_j(g,x) \psi_j(g,x). \tag{39}$$

Finally, noting that

$$\omega_j(g,x) = \sum_{k=j}^{J} \psi_k(g,x),$$

Equation (6) follows directly from Lemma 1. □

A.2 Lemma 1 and Its Proof

Lemma 1. For any constants $\{a_j, c_j\}_{j=0}^J$,

$$\sum_{j=0}^{J} a_j c_j = a_0 \tilde{c}_0 + \sum_{j=1}^{J} (a_j - a_{j-1}) \tilde{c}_j, \tag{40}$$

where

$$\tilde{c}_j \equiv \sum_{k=j}^{J} c_k. \tag{41}$$

Proof of Lemma 1. The proof is a summation-by-parts argument. Since $\tilde{c}_j$ is the tail

sum of the sequence $\{c_k\}_{k=j}^{J}$, we have

$$c_j = \tilde{c}_j - \tilde{c}_{j+1}. \tag{42}$$

Substituting this identity into the original sum gives

$$\sum_{j=0}^{J} a_j c_j = \sum_{j=0}^{J} a_j(\tilde{c}_j - \tilde{c}_{j+1}) \tag{43}$$

$$= \sum_{j=0}^{J} a_j \tilde{c}_j - \sum_{j=0}^{J} a_j \tilde{c}_{j+1}. \tag{44}$$

We now separate the first term in the first sum and re-index the second sum from j to $j-1$:

$$\sum_{j=0}^{J} a_j c_j = a_0 \tilde{c}_0 + \sum_{j=1}^{J} a_j \tilde{c}_j - \sum_{j=1}^{J} a_{j-1} \tilde{c}_j. \tag{45}$$

Collecting terms with the same $\tilde{c}_j$ yields

$$\sum_{j=0}^{J} a_j c_j = a_0 \tilde{c}_0 + \sum_{j=1}^{J} (a_j - a_{j-1}) \tilde{c}_j. \tag{46}$$

Hence, the claimed identity follows. □

A.3 Proof of Proposition 4

The first part of the proposition follows directly from Equation (6). If the weights satisfy the non-negative weights condition,

$$\omega_j(g, x) \geq 0 \quad \text{for all } j \geq 1, g, x,$$

and the level independence condition,

$$\omega_0(g, x) = 0 \quad \text{for all } g, x,$$

then the estimator β is a non-negative weighted average of treatment increments $\mu_j(g, x) - \mu_{j-1}(g, x)$. Therefore, β satisfies Definition WC.

To prove the converse, we argue by *contraposition*. Specifically, we show that if either the non-negative weights condition or the level independence condition fails, then there

exists a treatment response function μ such that

$$\mu_j(g,x) - \mu_{j-1}(g,x)$$

has the same sign for all $j \geq 1$ and all (g,x), while the resulting value of β has the opposite sign. This contradicts Definition WC and therefore shows that β cannot be weakly causal.

Step 1: Violation of level independence.

Suppose the level independence condition fails, but the non-negative weights condition holds. Then there exists some (g^*, x^*) such that

$$\omega_0(g^*, x^*) \neq 0,$$

while $\omega_j(g,x) \geq 0$ for all $j \geq 1$, g, and x.

Construct the treatment response function

$$\mu_j(g,x) = \begin{cases} \bar{\mu}, & (g,x) \neq (g^*, x^*), \\ \mu^*, & (g,x) = (g^*, x^*),\ j < j^*, \\ \mu^* + \Delta^*, & (g,x) = (g^*, x^*),\ j \geq j^*, \end{cases} \tag{47}$$

where $\bar{\mu}$, μ^*, and Δ^* are constants and $j^* \geq 1$ is arbitrary.

Interpretation of the construction. The treatment response is constant everywhere except for the group (g^*, x^*), where a single step change of size Δ^* occurs at treatment level j^*. Consequently,

$$\mu_j(g,x) - \mu_{j-1}(g,x)$$

is zero everywhere except when $j = j^*$ and $(g,x) = (g^*, x^*)$, where it equals Δ^*. Thus, the sign of the treatment increments is determined entirely by Δ^*.

Substituting the function (47) into Equation (6) and setting $\bar{\mu} = 0$ for simplicity yields

$$\beta = \omega_0(g^*, x^*)\mu^* + \omega_{j^*}(g^*, x^*)\Delta^*. \tag{48}$$

Now fix $\mu^* \neq 0$, so that $\omega_0(g^*, x^*)\mu^* \neq 0$. If $\omega_0(g^*, x^*)\mu^* > 0$, choose $\Delta^* < 0$ with sufficiently small magnitude such that

$$\omega_{j^*}(g^*, x^*)\Delta^* > -\omega_0(g^*, x^*)\mu^*.$$

Then

$$\beta = \omega_0(g^*, x^*)\mu^* + \omega_{j^*}(g^*, x^*)\Delta^* > 0,$$

while the treatment increment Δ^* is negative, violating Definition WC.

A symmetric argument applies when $\omega_0(g^*, x^*)\mu^* < 0$. In that case we choose $\Delta^* > 0$ sufficiently small so that $\beta < 0$. Hence Definition WC is violated.

Step 2: Violation of the non-negative weights condition.

Now suppose the non-negative weights condition fails. Then there exist (g^*, x^*) and j^* such that

$$\omega_{j^*}(g^*, x^*) < 0.$$

Using the same construction (47), set $\mu^* = 0$. Equation (6) then reduces to

$$\beta = \omega_{j^*}(g^*, x^*)\Delta^*.$$

Choosing any $\Delta^* > 0$ produces $\beta < 0$, even though all treatment increments are positive. This again contradicts Definition WC.

Conclusion.

We have shown that if either the non-negative weights condition or the level independence condition fails, then one can construct a treatment response function μ for which the estimator β violates Definition WC. Therefore, if β is weakly causal, both conditions must hold.

□

A.4 Summary Statistics

Table 3: Summary Statistics

	Grade				
Variable	K	1	2	3	All
Class size	20.3	21.0	21.1	21.3	20.9
	(4.0)	(4.0)	(4.1)	(4.4)	(4.1)
Percentile score avg. SAT	51.4	51.5	51.2	51.0	51.3
	(26.7)	(26.9)	(26.5)	(27.0)	(26.8)
Percentile score avg. BSF	NA	51.8	51.6	51.4	51.6
		(26.1)	(26.2)	(26.1)	(26.1)
Free lunch	.48	.52	.51	.50	.51
White	.67	.67	.65	.66	.66
Girl	.49	.48	.48	.48	.47
Age[a]	5.43	6.58	7.67	8.70	7.12
	(0.35)	(0.49)	(0.56)	(0.59)	(1.31)
Exited sample[b]	.29	.26	.21	NA	.43
Retained	NA	NA	NA	.04	NA
Percent of teachers[c] with MA+ degree	.35	.35	.37	.44	.38
Percent of teachers who are White	.83	.82	.79	.79	.81
Percent of teachers who are male	.00	.00	.01	.03	.01
No. of schools	79	76	75	75	80
No. of students	6,323	6,828	6,839	6,801	11,599
No. of small classes	127	124	133	140	524
No. of reg. classes	99	115	100	89	403
No. of reg/aide classes	99	100	107	107	413

Notes: This table reproduces the summary statistics reported in the appendix of Krueger (1999). Means are reported, with standard deviations in parentheses where available.
[a] Age as of September of the school year in which students are observed.
[b] The fraction that exited the sample in the next year for grades K–2; for All, it is the fraction that ever exited the sample.
[c] Teacher characteristics are weighted by the number of students in each teacher's class.